

INTERNAL · DRAFT

O2A-Synapse-Discovery

(PDiLL Enabled)

Technical Design Document — from process workflow to governed, executable SQL across enterprise datasources

PRODUCT	O2A-Synapse-Discovery — PDiLL Enabled
ENVIRONMENT	PDiLL (Production Data In Lower Environment) · part of O2A (Observe to Agent)
VERSION	0.1 — Draft
DATE	16 September 2026
OWNER	AJ (Ayyanar Jeyakrishnan) — Executive Director, AI Solutions Engineering
CLASSIFICATION	Internal — Wells Fargo. Handles PII and Production Data; access contained and audited.

1 Purpose & scope

O2A-Synapse-Discovery turns an uploaded process workflow into **validated, executable SQL** across many enterprise datasources, grounded on a machine-built metadata dictionary and a companion service-attribute catalog. It is the discovery-and-query layer of the broader **O2A (Observe to Agent)** program, delivered through the **Synapse Gateway** for controlled execution.

Because the product works with **PII and production data end to end**, it runs inside **PDiLL** — a gated, entitlement-controlled, fully audited environment. The governing principle is that production is *read only for metadata* while *query execution is confined to a lower (UAT) environment*, so sensitive data is discovered and modelled without being exposed or moved.

This document specifies the solution across four capabilities and the cross-cutting security and audit model, and is illustrated by four reference diagrams (Figures 1–4).

IN SCOPE

- Datasource onboarding (REST/API, SOAP, database) and a one-time metadata crawl.
- A metadata dictionary (MongoDB) with profiling, indexes, relationships, a join graph, and LLM-enriched semantics.
- Workflow-driven SQL construction, validation, and controlled Test execution in UAT.
- A SOAP/API service-attribute finder (MWS Attribute Finder) mapped to the dictionary.
- Identity, entitlement, containment, and Splunk audit throughout.

2 Solution overview

Four capabilities carry the product, over one shared security boundary and one shared knowledge layer:

#	CAPABILITY	RESPONSIBILITY	KEY STORE / IDENTITY
1	Onboarding Service	Connect and register a datasource (REST/API, SOAP, database); probe capability.	Global Service Account
2	Metadata Dictionary	One-time crawl — profiling, indexes, relationships, hashed 1,000-record sample; LLM-enriched semantics.	MongoDB
3	O2A Process → SQL Builder	Upload workflow → identify flow → resolve attributes → construct, validate, and Test SQL.	Synapse Gateway
4	MWS Attribute Finder	Ingest SOAP/API/screen metadata, index attributes, map service ↔ data.	Teradata / MWS

GUIDING PRINCIPLES

- **Production is metadata-read-only** — the crawl reads structure and hashed samples, never bulk production data.
- **Execution is UAT-only** — constructed SQL runs read-only in UAT via Synapse Gateway.
- **One entitlement** secures MongoDB and the onboarded datasources.
- **Separate service accounts** for crawl and execution — least privilege, separation of duties.
- **All access contained in PDiLL and fully audited via Splunk.**
- **Governed LLM and Prompt Playground are non-retaining** — no content logging, no training reuse.

3 End-to-end architecture

Figure 1 shows the full system: a single access boundary on top, the experience and agent layers, the knowledge layer, controlled execution, and shared governance services.

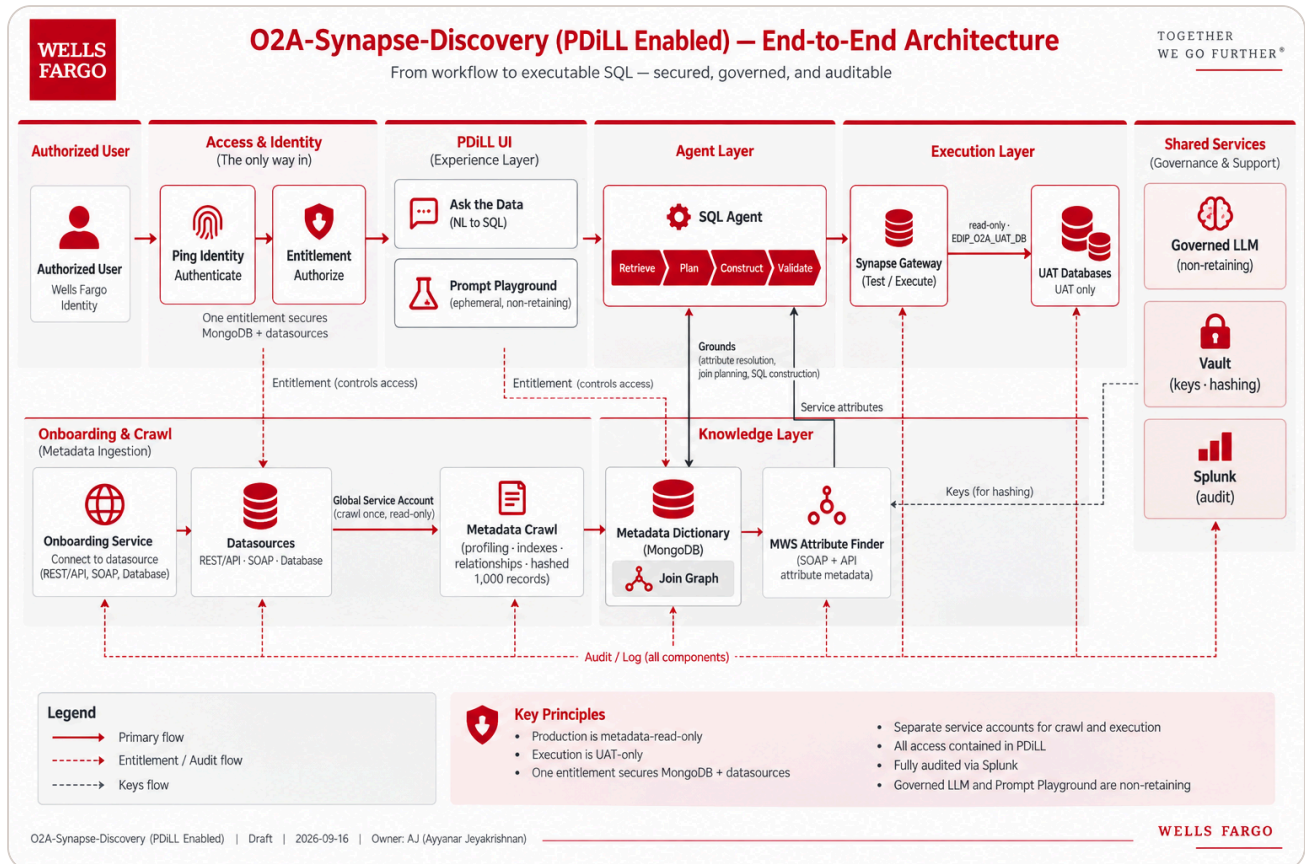


Figure 1. O2A-Synapse-Discovery (PDILL Enabled) — end-to-end architecture. From workflow to executable SQL, secured, governed, and auditable.

Reading the architecture

ACCESS & IDENTITY — THE ONLY WAY IN

Ping Identity authenticates the user (Wells Fargo SSO/MFA); an **Entitlement** check then authorizes access. A single entitlement secures both the metadata dictionary (MongoDB) and the onboarded datasources, so a user can never reach metadata about data they are not entitled to see.

EXPERIENCE LAYER — PDILL UI

Two surfaces sit behind the gate: **Ask the Data** (natural language → SQL) and the **Prompt Playground** (an ephemeral, non-retaining sandbox whose responses are shown once and discarded).

AGENT LAYER

The **SQL Agent** runs a deterministic loop — *retrieve* → *plan* → *construct* → *validate* — grounded on the dictionary and its join graph.

KNOWLEDGE LAYER

The **Metadata Dictionary (MongoDB)** holds schema and semantic metadata, relationships and join paths, business definitions and lineage, and hashed sample data with sensitivity tags. The **Join Graph** is the asset that lets the agent build correct cross-database joins. The **MWS Attribute Finder** contributes SOAP/API service-attribute metadata.

EXECUTION LAYER

Constructed SQL is executed as a **Test** step, read-only, in **UAT** only, through the **Synapse Gateway** using the `EDIP_02A_UAT_DB` service account. Production is never queried for data.

SHARED SERVICES

A **governed, non-retaining LLM**; **Vault** for keys and hashing; and **Splunk** for audit. Dotted lines in the figure show entitlement control, key distribution, and audit logging reaching every component.

4 Capability 1 — Onboarding Service

The Onboarding Service registers and connects a source as a first-class connector across three protocols — **REST/API**, **SOAP**, and **database**. On onboarding it verifies reachability and probes what the source exposes (endpoints or tables). Connection secrets are resolved from **Vault / AWS Secrets Manager**; the subsequent metadata crawl connects read-only using the **Global Service Account**.

The service is the front door for Capability 2: once a source is onboarded, its structure and a hashed sample are crawled once and loaded into the dictionary.

5 Capability 2 — Metadata Dictionary

A **one-time crawl** profiles each onboarded source and loads the result into **MongoDB** as the system of reference for SQL construction.

WHAT THE CRAWL CAPTURES

- **Data profiling** — data types, null percentage, distinct counts, value patterns.
- **Index details** — primary keys, unique and secondary indexes.
- **Relationships** — declared PK/FK plus inferred relationships, stored as a **join graph** with deterministic keys for cross-database joins.
- **Hashed sample** — a **1,000-record** sample, hashed (HMAC via Vault) so evidence is retained without keeping raw values at rest; carries **sensitivity tags**.
- **LLM-enriched semantics** — business names, descriptions, value/enum meanings, and lineage.

ONE-TIME ACTIVITY

Crawling is a **one-time** activity, not a standing re-crawl. The dictionary is thereafter maintained by editing metadata in place; a later refresh is a deliberate, re-authorized crawl. Because schemas may drift after the crawl, a lightweight

INFORMATION_SCHEMA check at query time (structure only, no re-authorization) is recommended to flag tables that have diverged from the dictionary.

6 Capability 3 — O2A Process → SQL Builder

Figure 2 shows the eight-step builder pipeline and the two supporting blocks — the dictionary that grounds it, and the external validation/execution path.

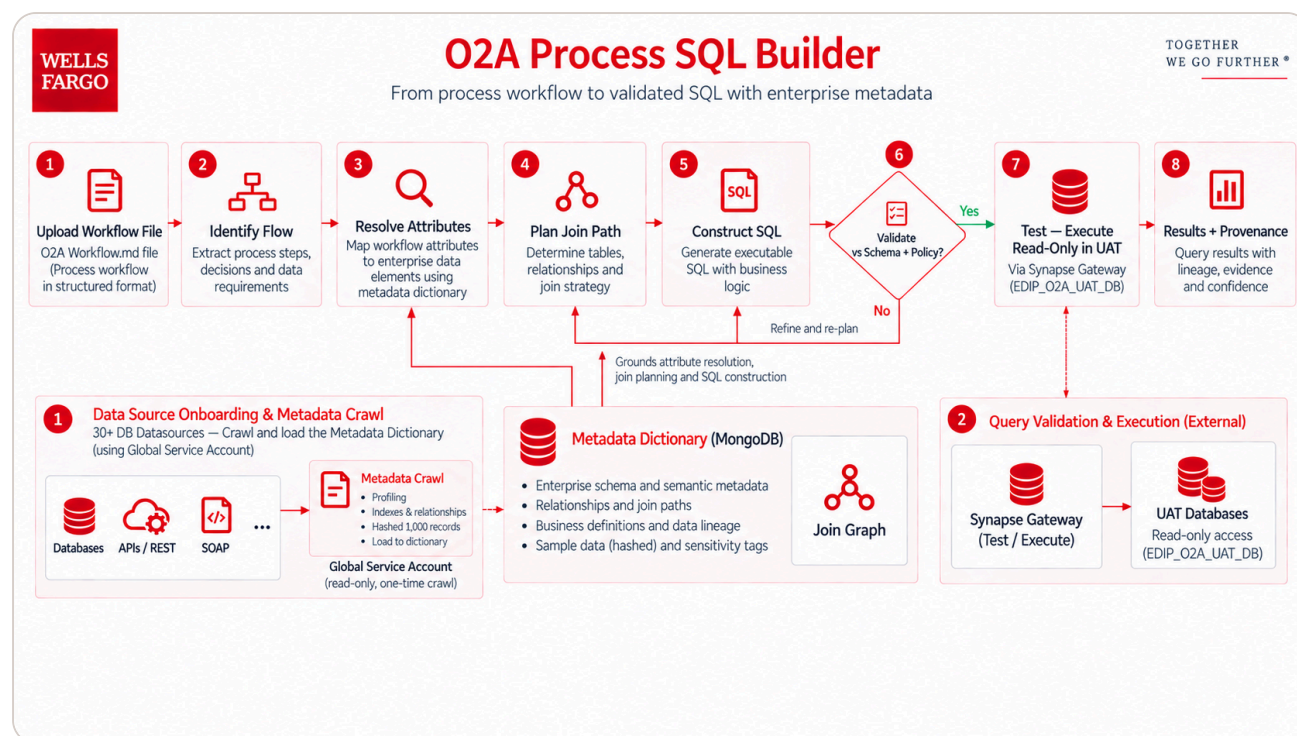


Figure 2. O2A Process → SQL Builder. From process workflow to validated SQL with enterprise metadata.

01 Upload Workflow File

An O2A workflow file (structured process definition) is uploaded as the builder's input.

02 Identify Flow

Extract the process steps, decisions, and data requirements — the ordered flow the SQL must realize.

03 Resolve Attributes

Map each step's data elements to enterprise attributes using the Metadata Dictionary.

04 Plan Join Path

Determine the tables, relationships, and join strategy — using the join graph, including cross-database keys. A flow may span one or more databases in a defined order.

05 Construct SQL

Generate executable SQL that encodes the flow's business logic.

06 Validate vs Schema + Policy

Check the SQL against the real schema and query policy. On failure, **refine and re-plan** (loop back to Plan Join Path); on success, proceed.

07 Test — Execute Read-Only in UAT

Run the validated SQL read-only in UAT via **Synapse Gateway** using `EDIP_02A_UAT_DB`.

08 Results + Provenance

Return query results with lineage, evidence, and a confidence signal.

The **Metadata Dictionary (MongoDB)** — enterprise schema and semantic metadata, relationships and join paths, business definitions and lineage, hashed samples and sensitivity tags — *grounds attribute resolution, join planning, and SQL construction* (steps 3–5). Onboarding and the one-time crawl (Global Service Account) populate it; validation and execution are external, through the Synapse Gateway to UAT.

7 Capability 4 — MWS Attribute Finder

Many enterprise processes reach data through services and screens, not just databases. The MWS Attribute Finder enriches **service** metadata — from WSDL, API specs, and application/mainframe screenshots — so users and the SQL Builder can find the right attributes and, where needed, construct the service call that retrieves them. Figure 3 shows the five stages and how they feed the O2A workflow.

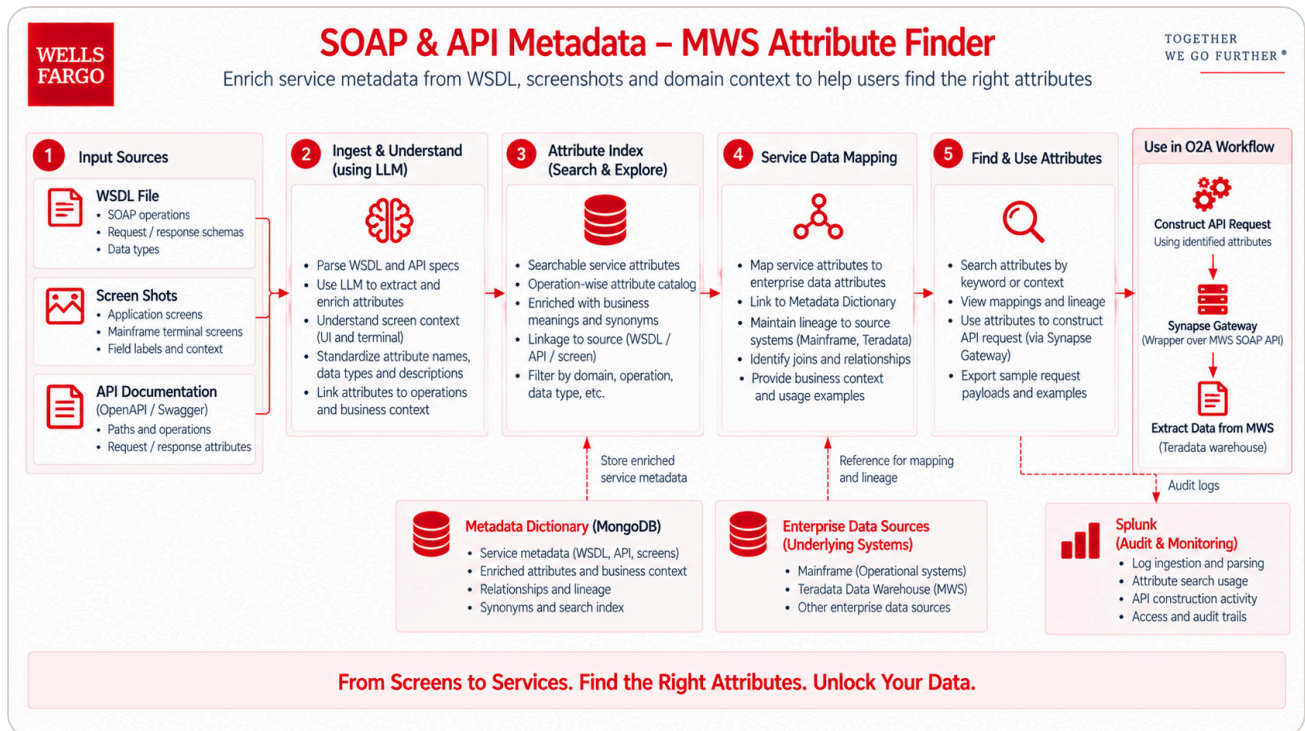


Figure 3. SOAP & API Metadata → MWS Attribute Finder. Enrich service metadata from WSDL, screenshots, and domain context to help users find the right attributes.

01 Input Sources

WSDL files (SOAP operations, request/response schemas, data types), application and mainframe terminal **screenshots** (field labels and context), and API documentation (OpenAPI/Swagger paths and operations).

02 Ingest & Understand (using LLM)

Parse WSDL and API specs; use the LLM to extract and enrich attributes; understand screen context (UI and terminal); standardize attribute names, data types, and descriptions; link attributes to operations and business context.

03 Attribute Index (Search & Explore)

A searchable, operation-wise attribute catalog enriched with business meanings and synonyms, linked back to its source (WSDL / API / screen), filterable by domain, operation, and data type.

04 Service Data Mapping

Map service attributes to enterprise data attributes; link to the Metadata Dictionary; maintain lineage to source systems (Mainframe, Teradata); identify joins and relationships; provide business context and usage examples.

05 Find & Use Attributes

Search by keyword or context, view mappings and lineage, use attributes to construct an API request (via Synapse Gateway), and export sample request payloads and examples.

USE IN THE O2A WORKFLOW

Identified attributes are used to **construct an API request**, sent through the **Synapse Gateway** (a wrapper over the MWS SOAP API), which **extracts data from MWS** — the **Teradata** data warehouse. Underlying systems include the **Mainframe** (operational systems), **Teradata (MWS)**, and other enterprise sources. Enriched service metadata is stored in the Metadata Dictionary, and all activity — attribute search, API construction, access — is logged to Splunk.

8 Security, PDiLL & audit

Figure 4 is the governance view: how a request is contained and traced from request to results, the guardrails on execution, the PDiLL access-security foundation, and the Splunk audit chain.

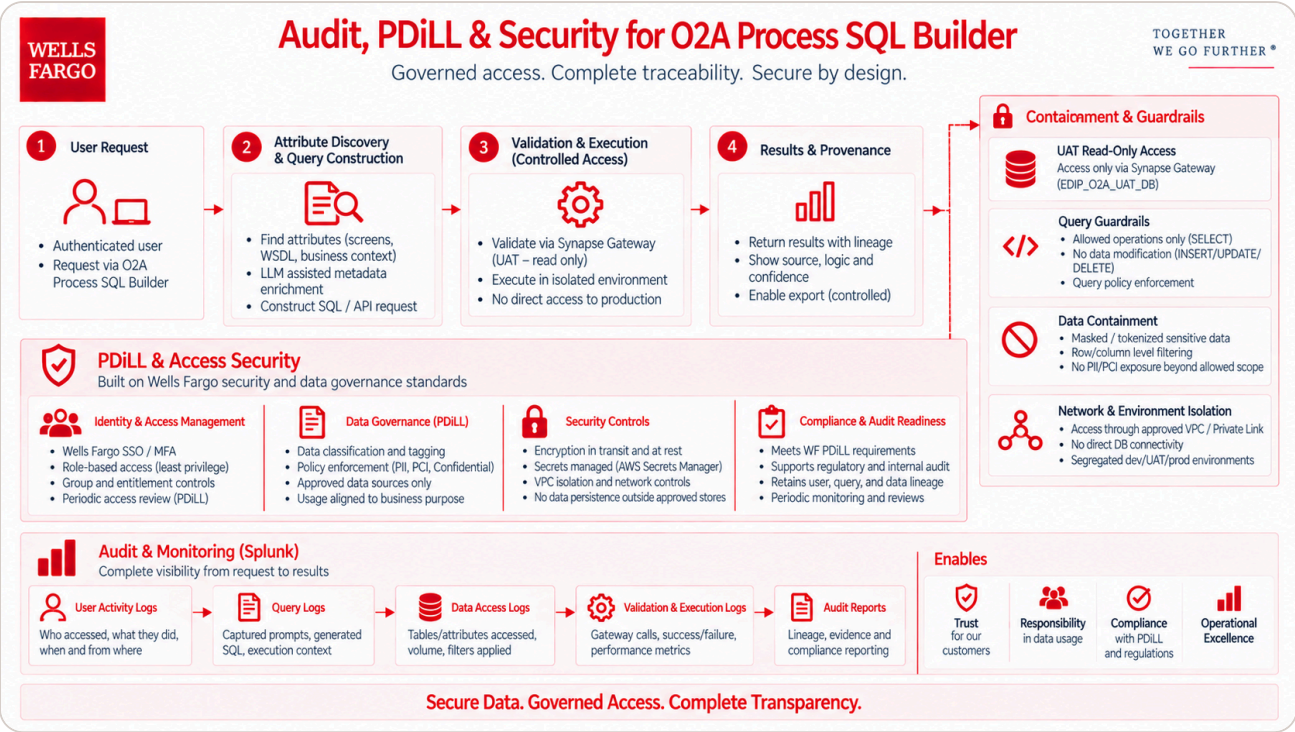


Figure 4. Audit, PDiLL & Security for O2A Process SQL Builder. Governed access, complete traceability, secure by design.

Containment & guardrails

- **UAT read-only access** — reached only via Synapse Gateway with `EDIP_O2A_UAT_DB`.
- **Query guardrails** — allowed operations only (`SELECT`); no `INSERT/UPDATE/DELETE` ; query-policy enforcement.
- **Data containment** — masked/tokenized sensitive data, row/column-level filtering, no PII/PCI exposure beyond the allowed scope.
- **Network & environment isolation** — access through an approved VPC / Private Link, no direct DB connectivity, segregated dev/UAT/prod environments.

PDiLL & access security

DOMAIN	CONTROLS
--------	----------

Identity & Access Management	Wells Fargo SSO / MFA (Ping Identity); role-based access (least privilege); group and entitlement controls; periodic access review (PDiLL).
Data Governance (PDiLL)	Data classification and tagging; policy enforcement (PII, PCI, Confidential); approved data sources only; usage aligned to business purpose.
Security Controls	Encryption in transit and at rest; secrets managed (AWS Secrets Manager / Vault); VPC isolation and network controls; no data persistence outside approved stores.
Compliance & Audit Readiness	Meets WF PDiLL requirements; supports regulatory and internal audit; retains user, query, and data lineage; periodic monitoring and reviews.

Audit & monitoring (Splunk)

Complete visibility from request to results: **user activity logs** (who accessed what, when, and from where), **query logs** (captured prompts, generated SQL, execution context), **data-access logs** (tables/attributes accessed, volume, filters applied), **validation & execution logs** (gateway calls, success/failure, performance), and **audit reports** (lineage, evidence, compliance reporting). Together these enable trust, responsible data usage, compliance with PDiLL and regulation, and operational excellence.

9 Connection identities (service accounts)

Two service-account identities separate discovery from execution — the account that builds the dictionary is never the account that runs queries.

PURPOSE	IDENTITY	ACCESS	PATH
Metadata crawl (build the dictionary)	Global Service Account	Read-only, one-time	Datasource → Metadata Dictionary (MongoDB)
SQL execution / Test	EDIP_02A_UAT_DB	Read-only, UAT only, SELECT	Synapse Gateway → UAT database

SEPARATION OF DUTIES

A compromised query path can reach only the lower (UAT) environment, never production. Production is read solely for metadata, under the Global Service Account; all row-level query execution is confined to UAT under EDIP_02A_UAT_DB.

10 Data model — catalog entry

Each dictionary entry combines authoritative structural facts, LLM-inferred semantics, hashed evidence, and provenance in one object:

```
// one column, fully described
{
  "catalog_id": "prod_cards.accounts.customer_id",
  "database": "prod_cards", "table": "accounts", "column": "customer_id",
  "structural": { "data_type": "varchar(20)", "nullable": false,
    "foreign_key": "customers.id", "row_count": 2384112 },
  "semantics": { "business_name": "Customer Identifier",
    "description": "Stable reference to the owning customer.",
    "is_join_key": true,
    "joins_to": ["prod_cards.customers.id", "prod_loans.borrower.cus
  "sensitivity": "PII",
  "sample_values": ["hmac:pII:9f2c...", "hmac:pII:4lad..."],
  "provenance": { "source_hash": "sha256:c7e0...", "sample_size": 1000 },
  "review": { "status": "approved", "owner": "cards-data-steward" }
}
```

11 NFRs & open decisions

NON-FUNCTIONAL REQUIREMENTS

- **Security** — encryption in transit/at rest; secrets in AWS Secrets Manager / Vault; VPC/Private Link isolation; no persistence outside approved stores.
- **Auditability** — every login and portal action logged to Splunk with retained user, query, and data lineage.
- **Least privilege** — distinct crawl and execution identities; SELECT-only execution.
- **Containment** — production metadata-read-only; execution UAT-only.

OPEN DECISIONS

- **Entitlement granularity** — one coarse grant vs. per-source entitlement enforced by the dictionary.
- **UAT ↔ production fidelity** — refresh cadence and masking, so UAT answers are trustworthy for their intended use.
- **Schema-drift detection** — query-time `INFORMATION_SCHEMA` check to flag divergence from the one-time crawl.
- **Catalog serving shape** — graph + embeddings for retrieval, so the agent can traverse join paths for multi-table SQL.

12 Glossary

TERM	MEANING
PDiLL	Production Data In Lower Environment — the gated, audited environment the product runs in.
O2A	Observe to Agent — the parent program bridging observed workflows to agentic execution.
Synapse Gateway	Controlled execution bridge; also a wrapper over the MWS SOAP API. Executes SQL/service calls read-only in UAT.
MWS	The Teradata data warehouse (and its SOAP API) that many services read from.
Metadata Dictionary	MongoDB store of schema, semantics, relationships, join graph, hashed samples, and sensitivity tags.
EDIP_02A_UAT_DB	Service account used for read-only SQL execution against UAT databases via Synapse Gateway.
Global Service Account	Read-only service account used for the one-time metadata crawl.